

A Probabilistic Cross-Shard Validator

Sofia Floody
The Johns Hopkins University

Abstract

The trilemma of blockchains states that security, scalability, and decentralization are not all currently achievable by blockchains. To improve one would mean to sacrifice another. The problem may be solved with sharding, a method that splits the blockchain network into smaller tasks. Sharding is being developed by networks, such as Ethereum. Sharding can improve latency, but only by sacrificing the security by having less validators or scalability by having more validators. This project introduces a parametric validator overlap model and shows that increasing overlap reduces the variance of adversarial concentration across shards, inducing a sharp phase transition in compromise probability in the critical region of overlap probability $p = .5-.7$.

1 Introduction

Current traditional blockchain implementations involve network nodes that contain the entire history of transactions. This method is common practice due to its high security and decreased complexity. The problem with this method is that it is inefficient as the history and transactions grow. A proposed solution is the sharding solution, a method meant to improve the scalability of blockchains by splitting the blockchain network into smaller shards that perform different tasks. This solution does decrease latency; however, it also decreases the security of the blockchain [16]. Each shard is separate and processes its own data, so there is limited cross-shard validation. A shard takeover risks local correctness and can affect other shards based on cross-shard dependencies [1].

A consensus mechanism is a fault-tolerant algorithm used in blockchain technology that is used to reach agreement among distributed nodes on the state of data in a decentralized manner [9]. Common types of consensus mechanisms include Proof-of-Work, Proof-of-Stakes, and Nakamoto consensus [19]. Consensus mechanisms are used to validate transactions. The mechanism we are focusing

on is Nakamoto. Nakamoto combines Proof-of-work, a computational method for proving validity, with a longest chain rule. The longest chain rule states that if forks occur in a blockchain, nodes follow the longest chain to reach a shared truth. A key feature of Nakamoto is the decentralization, being highly secure and Byzantine Fault Tolerant [15].

Validators that overlap between nodes can increase security as it will be more difficult for attackers to isolate one shard without being detected [12]. However, increasing overlap too much will make the model more inefficient and will start to resemble the classic blockchain model. Dynamically targeting high-risk shards with increased validator overlap solves this problem. A probabilistic cross-shard validator overlap can minimize shard takeover risk without drastically increasing latency. This project proposes a probabilistic model to analyze the effect of overlapping validators on the probability an attacker controls a shard and a small-scale simulation of randomly assigned validators and adversarial nodes using Ethereum-based architecture. The results will be used to compare a 0-overlap model, the probabilistic model, and a full overlapped model, determining the efficiency and security of the probabilistic model in the context of the existing models.

1.1 Claim

Let validators be assigned to shards with overlap probability p . As p increases, the variance of adversarial fractions across shards decreases monotonically. Consequently, the probability that any shard exceeds the adversarial threshold decays nonlinearly beyond a critical p .

2 Related Work

To limit memory cost, there are a variety of methods used for data compression and storage in sharding. Merkle trees store the root hash in the block header, allowing verification of transactions within a block. Nodes only require a Merkle

proof to verify a transaction, so shards can store partial transaction data. Erasure coding is another method for data compression. Erasure coding fragments, encodes, and distributes data across shards. The original data can be split into N fragments and encoded into K fragments stored in shards [7]. Any N fragments can reconstruct the data, so transactions can be verified with N fragments. This is secure since, if $K > 1$, it is nearly impossible to get source data from 1 shard. In a sharded system, transaction batches can be compressed using Merkle trees and these fragments can be distributed across shards using erasure coding. This allows validators to verify transactions without storing full datasets, still ensuring the original data can be reconstructed. Data can be defined as secure if a majority of the validators in the shard are not adversarial. 50% attackers in a shard will result in a corrupt shard, as defined by the Byzantine Fault Tolerance [4].

Ethereum is a blockchain that has begun employing sharding, including Layer 2 solutions that increase the throughput that inherit the security of the first layer of the blockchain. A popular Layer 2 solution is the rollup [2]. A rollup is a layer 2 crypto scaling solution that processes transactions off-chain, converts them to a single piece of data, and broadcasts them on a layer 1 blockchain [10]. There are two types of rollups, optimistic rollups and zero-knowledge (ZK) rollups. Optimistic rollups assume off-chain transactions are valid unless proven otherwise. ZK-rollups rely on cryptographic proofs before submitting proofs to Layer 1 that the off-chain computations are valid [3]. Rollups are used to process transactions faster since transactions are grouped together and they are processed off-chain: one single data point holds multiple transactions, and off-chain processing reduces the burden on the mainnet. This improves the scalability of blockchains.

Ethereum aims to employ danksharding. Danksharding creates a way for existing Ethereum layer 2 solutions to communicate. It aims to make rollups more efficient and introduces a way to process more data per block, so complex transactions will not slow down the network. Danksharding can allow the Ethereum network to process groups of transactions from the rollups without storing all their data immutably, further reducing burden on the mainnet, long term. However, to do this would take years, so they have decided to use proto-danksharding, an intermediate to danksharding.

The intermediate solution, as danksharding is being developed, is proto-danksharding. Proto-danksharding is a proposal that would provide the foundations to introduce danksharding to Ethereum. It introduces a new transaction type called “blob-carrying transactions” that use a new data type, “blobs” [11]. Blobs are binary large objects that temporarily store data. Currently, Ethereum stores all its

data on every node in the network and never deletes data. Although this is secure, it is computationally expensive and therefore inefficient. Blobs expire after 4096 epochs, reducing storage cost and, with blobs, each block can store an increasing amount of transaction data. Blob-carrying transactions carry a validity proof that allows the Ethereum chain to verify authenticity. Zero-knowledge proofs guarantee the EVM can view commitment to a blob, but not the data in the blob.

A disadvantage to Layer 2 solutions is that security shifts from full replication to probabilistic verification [6]. Sharding is a valuable way to increase efficiency of blockchains, but it is proving difficult to maintain high security. Existing overlapping proposals incorporate overlapping networks that enable all nodes in a shard to join other shards, enhancing security and improving communication efficiency [20].

3 Problem Formulation

The problem of cross-shard validator assignment involves a tradeoff between security and computational overhead in a sharded blockchain system. Let there be N validators and S shards. Each validator is assigned to a primary shard and may be assigned to additional shards according to the overlap probability p . Let f represent the global fraction of adversarial validators.

A shard, i , is assigned a set of validators V_i . Let q_i represent the fraction of adversarial validators in shard i . The system is compromised if the probability of shard compromise across all shards exceeds a certain fixed threshold.

There are two objectives: **Security Objective:** Minimize the probability that a shard is compromise. This is done in two different methods, one has a fixed threshold based on the Byzantine Fault Tolerance and one is done using the Nakamoto consensus.

Overhead Objective: Minimize the total number of validator-to-shard assignments.

These objectives conflict, as defined by the blockchain trilemma [13]. Increasing overlap reduces the variance of adversarial concentration within shards, improving security, but increasing computation overhead.

The goal is to identify an overlap probability O such that the security and overhead are optimized with an increased importance of security.

4 Methodology

4.1 Overlap Model

This project proposes a probabilistic cross-shard overlap model to balance both security and efficiency. Each validator is assigned to one shard to ensure a baseline level of security.

Each validator is assigned to one shard. For each validator and each shard that is not its initially assigned shard, assignment occurs independently with a probability of p . The value of p is tested through the following cases:

- $p = 0$: No overlap. Validators only cover their primary shard.
- $0 < p < 1$: Probabilistic Overlap. Validators are randomly assigned to more shards, increasing security without a great increase in overhead.
- $p = 1$: Full overlap. All validators cover all shards.

The span of the p value will offer comparisons for the probabilistic model to the baseline models of no overlap and full overlap, as well as in fine tuning the optimal probabilistic overlap.

4.2 Model A Parameters

The model of the sharded blockchain contains the following attributes:

- $N = 100$ Validators
 - This value is computationally efficient enough to be reasonable, also while providing enough diversity for random assignment effects.
- $S = 10$ Shards
 - This value is high enough to see trends, while not being so high to lose realism.
- $f = .4$ Fraction of adversarial validators
 - This is a high, but realistic fraction for adversaries. This falls in the Byzantine Fault Tolerance range, being high enough to stress-test the system, but not being so high that the simulation fails.
- $a = .5$ Fraction of attackers in shard required to compromise shard
 - This value is based on the Byzantine Fault Tolerance, stating that a majority of validators must be honest for a shard to maintain security, so 50% adversarial would compromise the shard [4]
- trials = 5000
 - Enough trials to ensure statistic reliability. Greater than previous value of 1000 trials.

These values were selected to create a relatively small-scale model to determine the compromise probabilities and overhead of different overlaps.

4.3 Model A Simulation Procedure

The procedure is based on the parameters and overlap model stated above. An assumption made is that a shard is compromised if the fraction of adversarial validators for that shard is ≥ 0.5 . This model represents a worst-case deterministic safety bound under Byzantine Fault Tolerance, acting as a baseline for comparison with a probabilistic consensus. Another assumption made is that if a shard is compromised, the trial is compromised.

1. Randomly select $(f * N)$ adversarial validators
2. Randomly assign 1 shard to each validator
3. Randomly assign extra shards to validators according to the overlap probability
4. Check if a shard is compromised if fraction of adversarial validators exceeds a
5. Return the ratio of compromised trials to total trials, the compromise probability
6. Repeat for all overlap probability values

4.4 Model B Introduction

Model A is a simple deterministic model that isolates the effect of overlap. Model A provides a sense of intuition. In order to create a more robust model that can test if this effect learned from Model A survives under strategic adversaries, adaptive defense, and probabilistic consensus. Model B is more realistic to the Ethereum network, the network these models are based off of. As a result, a secondary model, Model B, was introduced to recreate certain features.

Model B introduces a smarter adversarial system. The adversaries can attack certain shards with a probabilistic attack bias. Additionally, the honest validators can react to attacks based on defense strength and a suspicion threshold. These two properties will make the model more realistic to current blockchains and adversarial methods. Model B also introduces the Nakamoto consensus model, as opposed to relying on a fixed value for shard compromise. Model B calculates overhead in dynamically changing systems and is normalized. Model B redefines the compromise threshold to be 40% of shards compromised to yield a compromised trial. Both models offer valuable insight, so from this point forward the initial model will be referred to as model A and the secondary model will be referred to as model B.

4.5 Nakamoto Consensus

The initial model created assumes a shard is compromised if the percentage of adversarial validators exceeds a fixed value, 50%, following the Byzantine fault tolerance. This

assumption does not reflect proof-of-work and longest-chain consensus mechanisms, like the Nakamoto consensus that is used in Bitcoin. To create a more realistic probabilistic security model, the definition of shard compromise is extended to use the Nakamoto consensus framework. If adversaries control less than 50% of a shard, they can still corrupt a shard with some probability. Let q represent the fraction of adversarial validators in a shard and z represent the number of confirmation blocks. The probability that an adversary successfully overtakes the honest chain is approximated as $P_{attack}(q, z)$.

The probability decreases exponentially as number of confirmations increases. In this model including Nakamoto, a shard can be considered compromised probabilistically. For each shard, a compromise event occurs with the probability $P_{attack}(q, z)$ rather than deterministically without exceeding a threshold. This addition allows the model to represent time-dependent security guarantees, making it more accurate to real-world blockchain systems that rely on probabilistic finality.

The method for calculating Nakamoto is an approximation of Nakamoto behavior, not an exact formula.

4.6 Model B Parameters

Model B, in addition to the parameters consisting of model A, contains the following parameters.

- Attack Bias = .95
 - This controls strongly attackers target specific shards, with 1 being heavy attacks and 0 being more random.
- Target Shard Fraction = .5
 - This is the fraction of shards that attackers focus on attacking, modeling strategic attacks.
- Defense Strength = .6
 - This is the probability that an honest validator is added to a suspicious shard. A higher value is a stronger defense system, modeling adaptive security mechanisms.
- Suspicion Threshold = .4
 - This is the threshold for marking a shard as suspicious, so if .4 of validators are attackers, a defense is triggered.
- Confirmation Blocks = 10
 - This represents the number of blocks required for transaction finality in Nakamoto consensus mechanism. Higher values mean higher security.

4.7 Model B Simulation Procedure

1. Randomly select ($f * N$) adversarial validators
2. Adversarial validators select the number of target shards based on target shard fraction
3. Target shards are randomly selected
4. Each validator is initially assigned to shards, if validator is an attacker it has increased probability for attacking a shard already attacked based on the attack bias. An honest validator has a probability of p . All validators are then assigned to shards based on these previous probabilities
5. Empty shards are filled with validators
6. If the ratio of attackers in each shard over the number of validators is greater than the suspicion threshold, the shard is added to a set of suspicious shards
7. Honest validators can be reassigned to suspicious shards probabilistically based on the defense strength
8. Find compromised shards using the Nakamoto consensus mechanism
9. If the percent of compromised shards is 40% or greater, the trial is compromised
10. Repeat for all overlap probability values

4.8 Model B Adversary Model

The adversarial model in Model B is semi-strategic and coordinated, though it is still probabilistic. It is semi-adaptive, but not fully adaptive. Attackers choose their target shards at the beginning of each trial and do not update strategy in response to defense actions. This models a bounded rational adversary with partial coordination [14]. Real-world adversaries exhibit adaptive behavior, dynamically reallocating resources in response to defense strategies. However, in this model, fixing targets allows for the isolation of effects of probabilistic overlap without additional strategies. Although it is not the most realistic model, it is adequate for the purposes of this simulation.

Attackers choose a subset of shards to focus on based on the target shard fraction, then probabilistically attack shards based on the attack bias. The concentration for this is high, so attackers are likely to join target shards. This feature results in hotspots of vulnerability instead of uniform risk, like many real-world attackers would be. This probabilistic model is effective for this simulation, though would need to be updated for future work.

Honest validators can detect suspicious shards by the percentage of adversarial validators within a shard. The

honest nodes act more randomly and less coordinated. The contrast between the coordinated and concentrated attacks and the random uncoordinated behavior of honest nodes is realistic to actual blockchains. The system also has early warning, not just failure detection, as modeled by the suspicious nodes. This early reaction allows for defense, in which honest validators can reinforce suspicious shards based on the defense strength. This model creates an adaptive defense and load balancing, similar to reassigning validators and increasing redundancy on shards that are under attack. The attack success is then probabilistically calculated using the Nakamoto consensus.

Overall, this adversarial model has 3 key features that resemble real-world systems. First, it concentrates power, so attackers are not uniformly spread, increasing the chance of local takeover. Second, it creates asymmetry between the attackers and the honest nodes, with strategic attackers and random honest nodes, reflecting real systems. Third, it creates an attack vs defense dynamic. The concentrated attacks can be combated with detection and response.

A missing feature of this model is that attackers pick targets once and do not adapt after defense. Stronger adversaries will shift targets and react to defenses. This would require a less probabilistic model, and may not contribute much to the results of this paper, as the current model provides the features desired for the scope of this project.

4.9 Variable Refinement

In model A, troubleshooting was required to find a relationship that offered the most informative results. The final values outlined above were results of adjusting individual variables to optimize the results. For example, increasing f , the fraction of adversarial validators, decreases the security. If that value is too high, we cannot obtain information because all shards are always compromised. Essentially, my goal was to have fully overlapped have near 0 security risk and no overlap a high risk to visualize the points between. I landed on .4, which is also in a reasonable frame using previous data. This fraction is typically between .33 and .5 [6], so .4 is a reasonable value. The other value that required refinement was a , the fraction of adversarial validators in a shard to compromise the shard. After some adjustment, in the end I decided to maintain this value at .5, the standard value for sharded models. In development of model B, I wanted to keep these previous values stable for continuity purposes. As a result, I was required to adjust the new parameters as needed, as well as the percentage of compromised shards that yields a compromised trial. I was looking for the optimal set of values to create an s-curve of compromise probability that spanned from 0-1.

4.10 Result Generation

4.10.1 Model A

Security: The effect of overlap probability on security was determined by counting the number of shards compromised and dividing by the total number of trials per overlap probability. This value gives the probability of a trial being compromised over 5000 trials. Increased trial numbers increases the reliability of the results. Originally, 1000 trials were run, but switched to 5000 to increase reliability. This way we can compare the compromise probability per overlap probability.

Overhead: The effect of overlap probability on overhead was determined by counting the number of assignments each validator was assigned to. This value was divided over the total number of trials

4.10.2 Model B

Security: The compromise probability is calculated as the number of compromised trials over the total number of trials. A compromised trial is a trial that has $\geq 40\%$ compromised shards.

Overhead: Overhead is calculated by summing the total number of validators assigned, including when validators are added repeatedly and shards grow dynamically. This value is then normalized by N , the total number of shards.

4.10.3 Pareto Front

The Pareto Front is the set of optimal trade-off solutions in multi-objective optimization [18]. This is used to find the optimal overlap that maximizes security and minimizes overhead. A single-objective optimization is done following a Pareto Front model, but including a weight towards security.

The Pareto Front is approximated via a weighted scalarization of non-dominated solutions. Security is favored in this calculation to represent the importance of security in blockchain systems [5]. This represents the optimal overlap because it finds the best security value for the overhead before the tradeoff becomes too great.

5 Results

In addition to the deterministic Byzantine model, results can be interpreted under the probabilistic Nakamoto consensus framework. Using this framework, shard compromise is no longer binary based on the strict adversarial threshold of 50%. Instead, it reflects the probability of a successful adversarial chain reorganization. This results in smoother transitions in compromise probability and highlights the importance of confirmation depth in security.

5.1 Compromise Probability

5.1.1 Model A

Based on model A, at overlap probability $p=0$, the compromise probability is high, at 99.4%. With overlap probabilities at $p=.9$ and $p=1$, the compromise probabilities were 0%. The compromise probability decreases non-linearly with the overlap probability. This means that small increases in p early on do not affect the security very much. The mid-range of p values of .4-.7 have the most rapid security improvement. This is likely the range of overlap that is optimal for the model. The threshold region of .5-.6 security improves rapidly. From $p = .5$ the compromise probability is 25.6%, at overlap probability $p=.6$ the compromise probability is 9.3%, and at $p=.7$ the compromise probability is 1.4%. This extreme drop suggests that this region demonstrates great security benefit from increasing overlap by this amount. From .5 to .7, the security is approximately 18 fold better at .7 than .5. This appears to be a sharp nonlinear transition. After overlap probability $p=.7$ the security gains diminish and compromise probability is already extremely low, making the overhead cost not worth the marginal additional security benefits.

5.1.2 Model B

The compromise probability at overlap probability $p=0$ in Model B is similar to that in Model A, being 98.6%. The compromise probability at $p = 1$ does not reach 0, though it does reach 6.3%. This value is more realistic than Model A, as a system is very unlikely to be 100% secure, so a model with 6% compromise probability is likely more trustworthy than one with a 0% compromise probability, as the 6% is more likely to model a real blockchain. The compromise probability decreases non-linearly, in an s-curve shape. There is a small decrease in compromise probability from $p=0$ to $p=.2$, and then a more rapid decrease from $p=.2$ to $p=.7$, the sharp nonlinear transition, then starts to level out and decreases less rapidly from $p=.7$ to $p=1$. The benefit from increase overlap at high values, such as from $p=.6$ to $p=1$ is greater than that in Model A, though may not be worth the overhead tradeoff.

5.2 Overhead

5.2.1 Model A

Based on model A, the overhead starts at 10 at $p=0$ and increases to 1000, the maximum at $p=1$. Each shard inherently has one validator in this model, so the minimum overhead must be 10. At an overlap of 1, each validator is assigned to each shard, so the overhead is $10 \times 100 = 1000$. This is a 900% increase. Although $p=1$ is fully secure, with the increased overhead, it negates the scalability benefits of sharding. This reinforces the need for a probabilistic model, as full overlap is extremely inefficient for marginal security benefits. Lin-

ear increase means the cost of increasing security is very predictable.

5.2.2 Model B

With the new overhead calculation for Model B, the values are very different from Model A. The overhead for 0 overlap, the is 4.468. Strangely, the overhead for $p=.1$ is slightly lower than $p=0$ at 4.149. This is the only value to decrease in overhead from increasing overlap. The values increase in a linear fashion after the first dip. The overhead at $p=1$ is 8.099, so there is an 80% increase from the overhead at $p=0$ to $p=1$. The new method for computation still yields a linear increase. The full overlap model is greatly inefficient, and while the 0 overlap is efficient, does not have the security needed.

5.3 Compromise vs Overhead

5.3.1 Model A

Considering both compromise probability and overhead together, the results highlight the tradeoff between security and overhead. Full overlap eliminates the compromise risk, but is the most computationally expensive. No overlap has low computational cost, but extremely high compromise probability. With overhead increasing linearly, the region of $p=.5-.7$ seems to be optimal. The security benefit from .5 to .7 is 18 fold, while the overhead detriment from .5 to .6 is .7. This security benefit greatly outweighs the computational cost of increasing overlap.

5.3.2 Model B

Model B produces an s-curve, with fast nonlinear decay in compromise probability occurring around overlap probability $p=.2-.7$. This is a similar, but broader range than that of Model A. The Model B curve is less distinct and sharp, being more flat. The compromise probability vs overhead relationship for Model B is defined using the previously defined Pareto Front. The compromise probability and overhead that are found to be optimal using the calculations are .235 and 6.779, respectively. These values correlate to 70% overlap.

5.3.3 Model A vs Model B

Model A and Model B seem to yield similar results in terms of the optimal compromise probabilities. Model A seemed optimal in the range of $p=.5-.7$. Model B seemed optimal in the range of $p=.2-.7$, overlapping with the range Model A found optimal. Using the Pareto Front in Model B, it seems that 70% overlap is ideal for balancing security and overhead, with security being prioritized. The security in Model A at $p=.7$ is far less compromised than in Model B, which speaks to the realism of Model B over A. It is unlikely that a real model would be that secure at that overlap, so this goes to

show that Model B is more realistic to actual Blockchains. Overhead is not directly comparable across models as the overhead calculation has shifted from Model A to Model B.

Overlap Probability	Compromise Probability	Overhead
0.0	0.994	10.00
0.1	0.976	100.06
0.2	0.884	199.99
0.3	0.699	299.75
0.4	0.466	400.30
0.5	0.256	499.89
0.6	0.093	599.82
0.7	0.014	700.29
0.8	0.001	799.87
0.9	0.000	899.77
1.0	0.000	1000.00

Table 1: Model A Results

Overlap Probability	Compromise Probability	Overhead
0.0	0.986	4.468
0.1	0.965	4.149
0.2	0.899	4.582
0.3	0.791	5.018
0.4	0.644	5.458
0.5	0.476	5.898
0.6	0.333	6.340
0.7	0.235	6.779
0.8	0.146	7.221
0.9	0.096	7.659
1.0	0.063	8.099

Table 2: Model B Results

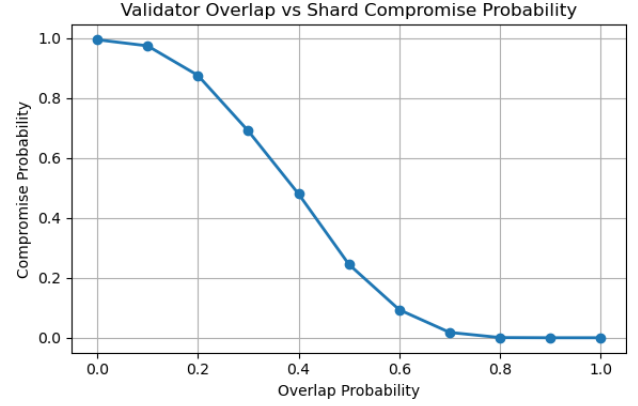


Figure 1: Overlap vs Security Graph Model A

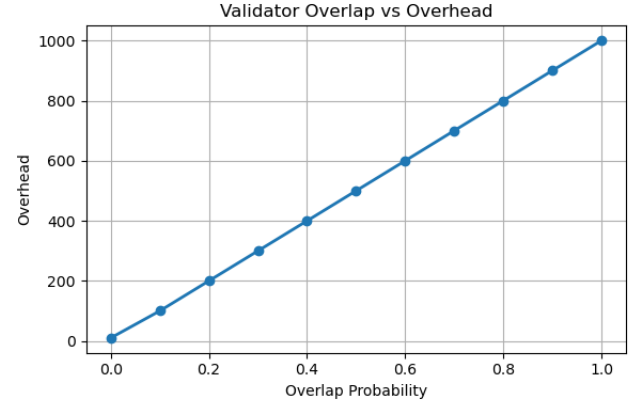


Figure 2: Overlap vs Overhead Graph Model A

6 Discussion

These results show that a probabilistic cross-shard validator overlap can significantly improve shard security without incurring the high computational cost of full overlap. All validators do not need to participate in all shards. Moderate overlap probabilities are nearly as secure as fully overlapped models. Additionally, models that are not fully overlapped, but nearly so, such as the models with $p=.8$ and $p=.9$ have essentially the same security benefits as the full overlapped model with less overhead. Even if one wanted a fully secure model, these options would be computationally less costly than the fully overlapped model.

Cross-shard validators increases the validation requirements in sharding compared to validators that do not cross shards. As overlap probability increases, adversarial validators are less likely to gain concentration within a single shard, making it difficult for them to achieve either the 50% baseline

or the Nakamoto consensus necessary to compromise a shard. The likelihood of reaching this threshold is far more difficult with higher overlap probability. This method of redistribution shows greatly in the mid-range of overlap probabilities. This range shows that small changes in overlap lead to great decreases in compromise probability with low additional computational cost. Full overlap eliminates compromise risk, though it introduces inefficiency. The goal is to find a scalable and secure solution. Moderate overlap shows to provide a balance between security and computational cost.

Based on the results for Model B, it seems that the optimal overlap is 70% overlap. This value also aligns with the results from Model A. Though Model A is simpler than Model B, it is noteworthy that the more simple model was able to produce similar results to a model that is more complex.

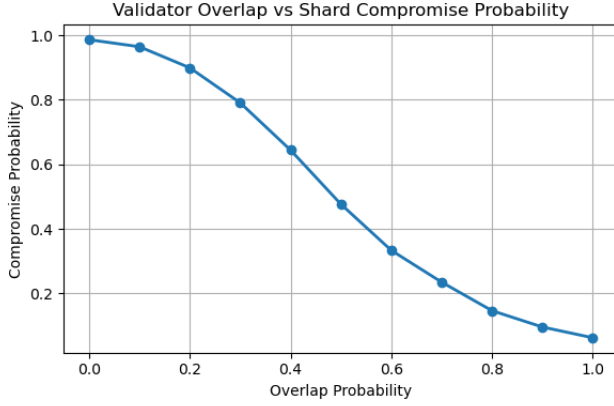


Figure 3: Overlap vs Security Graph Model B

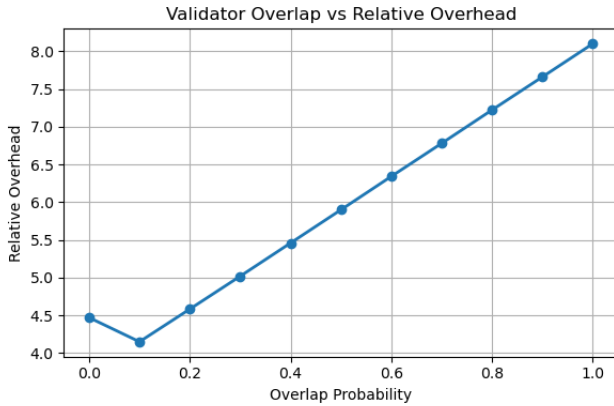


Figure 4: Overlap vs Overhead Graph Model B

6.1 Blockchain Design Applications

The results are relevant to sharded blockchain systems, such as the ones used in Ethereum. These shards do not need to be fully isolated, disconnected values, and they do not need to be fully replicated. A hybrid, probabilistic overlap can offer strong security without an unreasonable amount of overhead. Systems, like the Ethereum system, assign validators to subsets of shards, but they have limited bandwidth. Tuning the percentage of overlap can reduce the risk of shard compromise without a full overlap model. Current focus on probabilistic verification in Layer 2 and sharded architectures aligns well with this probabilistic overlap model. Increasing overlap and redundancy can improve security risks without creating a fully overlapped model, offering both security and efficiency. The results show possibility for a future adaptive system where overlap is dynamically increased or decreased in response to perceived threat risks. Certain markers, like adversarial activity or suspicious behaviors, can activate increased validator overlap, further improving security, witnessed in Model B.

6.2 Probabilistic Interpretation

In a no-overlap system, each shard is assigned a small and isolated subset of validators. With many adversaries, random fluctuations can cause the shard to exceed the 50% compromise threshold. As overlap increases, validators are distributed across multiple shards, increasing the sample size of validators per shard. This means there will be less variance in adversaries per shard, making unlikely outcomes, like majority adversaries, less likely to occur. So, probabilistic overlap reduces variance, normalizing the shard security across the network and making it more stable. This is why there is a sharp decline in compromise probability in the mid-range of p values. This indicates the possibility of a critical overlap threshold where adversarial effects collapse greatly. The system is in a more stable validator distribution.

6.3 Consensus Mechanisms

Initially, in Model A, there was no consensus mechanism, rather a shard was compromised if at least 50% of validators were adversarial. This comes from the Byzantine Fault Tolerance. This is not a process for reaching agreement, such as a consensus mechanism, but is a condition on safety. This makes Model A a deterministic failure model as opposed to a consensus protocol. This was useful for its use, allowing for first-order approximation and worst-case safety boundaries. This was an important step in development, creating a threshold for the future, more realistic model.

Including a Nakamoto consensus in Model B upgraded the realism because it captures partial adversarial control and models attack probability over time. It is not binary, like Model A, and is probabilistic. It reflects how real

systems, like Bitcoin and Ethereum have historically behaved. The thresholds and parameter tuning from Model A was significant to the development of Model B.

6.4 Scalability Implications

These results can be extended to larger-scale blockchain networks. This model uses only 100 validators and 10 shards, but real-world systems, like Ethereum, have significantly larger validator sets. As the scale increases, the probabilistic effects shown in this simulation should be more pronounced. As scale increases, the distribution of validators across shards should be more stable since large numbers tend to follow statistical distributions more. The benefits of probabilistic overlap could be greater at larger scale, since variance in adversarial concentration within shards decreases. Lower overlap probabilities could offer secure systems for larger networks that are not visible on this smaller scale.

Alternatively, increasing system size also introduces computational costs. In this model, overhead is represented by validator assignments, but real-world implications have network latency, limits on bandwidth, and other computational costs. These factors would shift the optimal overlap range. Because of this, more realistic, large-scale environments should be tested, as discussed in the Future Work section.

6.5 Security Implications

These results suggest an underlying relationship between overlap probability and the probability of shard compromise. As overlap increases, each shard is assigned to more validators and does so in a more uniformly distributed way. This decreases the chances of a shard having a disproportionately high percentage of adversarial validators.

The probability of a shard being compromised is the probability that a random subset of validators exceeds the compromise threshold for adversaries. Increasing overlap changes these subsets, introducing redundancy across shards to reduce the chance of high proportion of adversaries. This explains why the compromise probability decreases so rapidly in the mid-range of the overlap probabilities.

Under Nakamoto consensus, security is not strictly determined by the 50% adversarial threshold in a shard. Instead, security is contingent on both adversarial fraction and time, as is measured in confirmation blocks. Shards with less than 50% adversarial validators can be vulnerable to corruption with small confirmation depth, while large confirmation depth can secure shards under moderate adversarial proportions. This reinforces the importance of the probabilistic overlap model. Overlap can prevent deterministic takeover and

exponentially reduce the probability of a successful attack under the Nakamoto consensus by reducing adversarial fraction within shards

6.6 Parameter Sensitivity

The results are sensitive to the parameters listed in the methodology section. Slight changes to even one parameter can drastically alter the results. The parameters selected were based on both existing values and concepts, as well as fine tuned to avoid degenerate cases, such as always compromised or always safe. Parameters such as adversarial fraction and number of shards can have great impacts. For example, increasing adversarial fraction shifts the compromise curve upward, making the system less secure and requiring higher overlap to maintain security. Increasing number of shards reduces the number of validators per shard, increasing the likelihood that a shard can be taken over, making it more vulnerable. This suggests that the optimal overlap is not fixed, but depends on system scale, threat assumptions, and many other factors that have been fine tuned for this particular system to ensure measurable variance across p .

6.7 Considerations for Increasing Realism

In real-world applications, a probabilistic-cross shard overlap would need to coordinate between validator assignment mechanisms and network resource limits. Validators have limited computational power, so they cannot be assigned to an infinite number of shards. They must be balanced and proportionate to their ability to process and verify transactions. More dynamic models that require validators to join and leave the network could be computationally expensive to maintain the overlap distribution.

Validators in blockchain networks are rewarded for assignment to shards, so the mechanisms would have to reward validators for increased shard verification. Without this, the security benefits would be diminished.

Another consideration is incentive design. Validators in real blockchain systems are typically rewarded for participation, so mechanisms would need to ensure that validators assigned to multiple shards are compensated appropriately for increased workload. Without proper incentives, uneven participation could undermine the intended security benefits of probabilistic overlap.

The current adversarial model is semi-strategic, but still probabilistic. This is not fully optimal. Future implementations should have a game-theoretic attacker that is not probabilistic to make the model follow real-world principles more closely.

6.8 Comparison to Existing Sharding Architectures

Existing blockchain systems employ different approaches to validator assignment. Ethereum has a proposed sharding model in which validators are randomly assigned to groups that regularly rotate, reducing the likelihood of sustained adversarial control [8]. These assignments are typically not overlapping. The probabilistic overlap model introduces some amount of redundancy across shards. This creates a mix between committee-based security and full replication. Probabilistic overlap reduces the risk of single-shard isolation attacks by introducing cross-shard validation.

Layer 2 systems, such as rollups, rely on data availability and fraud proofs rather than validator overlap [17]. The proposed model in this paper operates at the validator assignment layer, making it a possible addition to these approaches, not a replacement. This suggests that probabilistic overlap can be integrated into existing architectures as an additional security layer, not an entirely new design.

6.9 Novelty

This model introduces a continuous overlap parameter with a parametric continuum between fully isolated shards and fully replicated blockchains. Most existing models have a fixed committee size and fixed assignment rules, but this model introduces more dynamic parameters.

Results show there is a nonlinear phase transition in validator overlap in the region between $p=.5$ and $p=.7$. This section is where adversarial concentration collapses rapidly. This region characterizes the security requirements of a sharded blockchain, showing the amount of overlap required to ensure blockchain security.

Security improvements come from variance suppression, not just redundancy. Overlap reduces the variance of adversarial concentrations across shards.

The Byzantine threshold model and Nakamoto models can act as deterministic and probabilistic models, respectfully. This paper focuses on both models, comparing the two and the effects of validator overlap on them. It is clear that overlap effects persist under both deterministic and probabilistic consensus.

6.10 Limitations

There are a variety of limitations in this study. The simulation scale is relatively small, with only 100 validators. The overhead model can still be improved, though it was improved from Model A to Model B. The overhead model in Model B is

still relatively simple and results in a linear growth as overlap increases, a feature that is not likely to occur in more complex models. Finally, model A and B allow unlimited overlap for validators, which is not entirely realistic for deployed systems.

To address some of these limitations, future models should limit the maximum number of shards per validator to better reflect real-world systems. This limits validator workload while still preserving overlap. Limiting this value will likely decrease overhead but weaken security. This would mean that the overlap necessary to find the optimal security may be slightly higher.

6.11 Future Work

Future work would include limiting the number of shards a validator can be assigned to. This is a big piece that is missing from the puzzle, as it is unrealistic to models for validators to be able to assign to any amount of shards. Future work could also scale up the simulation to model larger networks. Future work would include implementing more strategic adversarial nodes. These methods would increase the reliability and complexity of this model, making the results more meaningful to real-world applications.

7 Conclusion

This work presents a probabilistic cross-shard validator overlap model that balances security and computational efficiency in sharded blockchain systems. This model introduces a tunable overlap parameter that allows for a balance between isolated shards and a fully overlapped model.

The simulation produced shows that overlap has a non-linear effect on system security. In both Models A and B, there is a critical transition region that emerges between overlap probabilities $p=.5$ and $p=.7$. Small increases in overlap result in large reductions in compromise probability. This feature, tied with the linear overhead model, makes this range optimal, as security exponentially increases with a constant increase in overhead. Model B approximates 70% as the optimal tradeoff point for overlap, while prioritizing security over overhead.

These results show that full overlap is not necessary to create a secure system. Moderate probabilistic overlap can provide near-equivalent protection against shard takeover while reducing computational overhead. Increased redundancy does not necessarily scale linearly with security, highlighting the importance of probabilistic design in distributed systems.

Incorporating Nakamoto consensus as a probabilistic

consensus framework allows for a more realistic representation of adversarial success. This approach captures the time-dependent, probabilistic nature of blockchain security, making the proposed overlap strategy more robust.

Overall, this work demonstrates that probabilistic cross-shard validator overlap is a reasonable and efficient mechanism for improving security in sharded blockchain systems. These results offer a foundation for future adaptive systems that use dynamic overlap based on network conditions, adversarial behavior, and resource constraints.

This work highlights the importance of probabilistic design in distributed systems. These systems should not rely on strict thresholds or full redundancy, as these do not offer the security or the efficiency required of a blockchain. Carefully tuned randomness can provide strong guarantees with reduced cost. This principle could extend beyond blockchain systems to other distributed architectures.

References

- [1] COINBASE. What is crypto sharding and how does it work?, 2026.
- [2] COINBASE. What is danksharding and the difference with proto-danksharding?, 2026.
- [3] DYADE, A. M., DYADE, M. A., AND SHINDE, S. M. Cost optimization in layer 2 rollups via EIP-4844: A gas efficiency and economic analysis. *IET Blockchain* 5, 1 (Jan. 2025).
- [4] FINST. What is the byzantine generals problem?, 2025.
- [5] GIAGKIOZIS, I., AND FLEMING, P. J. Pareto front estimation for decision making. *Evol. Comput.* 22, 4 (2014), 651–678.
- [6] JAMES, A. What are the advantages and disadvantages of layer 2 scaling solutions?, 2023.
- [7] KHEMANI, V. Erasure coding for the masses, 2020.
- [8] KUDZIN, A., TOYODA, K., TAKAYAMA, S., AND ISHIGAME, A. Scaling ethereum 2.0s cross-shard transactions with refined data structures. *Cryptography* 6, 4 (Nov. 2022), 57.
- [9] LASHKARI, B., AND MUSILEK, P. A comprehensive review of blockchain consensus mechanisms. *IEEE Access* 9 (2021), 43620–43652.
- [10] LEDGER. What are blockchain rollups?, 2022.
- [11] LEDGER. Danksharding and proto-danksharding explained, 2025.
- [12] LI, X. Security analysis of sharding in blockchain with pbft consensus. *ACM Digital Library* 220, 103748 (2022).
- [13] NAKAI, T., SAKURAI, A., HIRONAKA, S., AND SHUDO, K. The blockchain trilemma described by a formula. In *2023 IEEE International Conference on Blockchain (Blockchain)* (2023), pp. 41–46.
- [14] RABIEINEJAD, E., YAZDINEJAD, A., PARIZI, R. M., AND DEGHAN-TANHA, A. Generative adversarial networks for cyber threat hunting in ethereum blockchain. *Distrib. Ledger Technol.* 2, 2 (June 2023), 1–19.
- [15] SOMPOLINSKY, Y., WYBORSKI, S., AND ZOHAR, A. PHANTOM GHOSTDAG. In *Proceedings of the 3rd ACM Conference on Advances in Financial Technologies* (New York, NY, USA, Sept. 2021), ACM.
- [16] TEAM, T. I. What is sharding? purpose, how it works, security, and benefits, 2024.
- [17] THIBAUT, L. T., SARRY, T., AND HAFID, A. S. Blockchain scaling using rollups: A comprehensive survey. *IEEE Access* 10 (2022), 93039–93054.
- [18] TUŠAR, T., AND FILIPIĆ, B. Visualization of pareto front approximations in evolutionary multiobjective optimization: A critical review and the prosection method. *IEEE Trans. Evol. Comput.* 19, 2 (Apr. 2015), 225–245.
- [19] WANG, W., HOANG, D. T., HU, P., XIONG, Z., NIYATO, D., WANG, P., WEN, Y., AND KIM, D. I. A survey on consensus mechanisms and mining strategy management in blockchain networks. *IEEE Access* 7 (2019), 1–1.
- [20] YU, B. Overshard: Scaling blockchain by full sharding with overlapping network and virtual accounts. *ScienceDirect* (2023).